

Notes: Garmaise and Moskowitz (RFS, 2004)

Confronting Information Asymmetries: Evidence from Real Estate Markets

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1 Big picture

- **Question.** How important are asymmetric information problems in commercial real estate markets, and how do market participants respond to them through trading patterns and financing choices?
- **Setting.** The paper studies U.S. commercial real estate transactions from January 1996 to March 1999 using the COMPS database. The usable sample contains 10,351 property sales across Nevada, Massachusetts, Maryland, Virginia, Texas, Illinois, and Colorado.
- **Core challenge.** Information effects are hard to test directly because most empirical proxies for information quality are endogenous. A positive association between trading behavior and an “information proxy” could simply reflect local market conditions, local wealth, property heterogeneity, or institutional differences in financing.
- **Main conceptual contribution.** The paper introduces a direct and plausibly exogenous information measure: the quality of property tax assessments in a local jurisdiction. Regions with less accurate public assessments are treated as high-asymmetric-information environments.
- **Main hypotheses.** The paper tests three broad classes of implications:
 - **Limited participation:** distant buyers should avoid trading with better-informed locals, especially where information asymmetry is severe.
 - **Selective offering and market segmentation:** properties with short histories should be harder to sell in high-information-problem environments, and informed brokers should preferentially trade with each other.
 - **Financial structure:** seller financing should rise and bank financing should fall when information problems are severe, if financing is used to mitigate adverse selection.
- **Main empirical design.** The authors regress buyer distance, property age, broker-to-broker trading, financing outcomes, and capitalization rates on the coefficient of dispersion (COD) from state assessment ratio studies, plus rich controls for property type, local price heterogeneity, price growth, and geography.
- **Main baseline result.** Buyers are substantially more local in high-COD places. The paper summarizes this as: a one-standard-deviation increase in COD brings buyers more than 70 km closer to properties, and buyers in high-COD areas are more than 90 km closer than buyers in low-COD areas.
- **Key heterogeneity result.** The local-buyer effect is strongest for younger properties, and brokers are especially likely to trade with other brokers when COD is high.
- **Key contrast result.** The evidence for limited participation, selective offering, and broker market segmentation is strong. The evidence that financing structure solves the information problem is weak and mixed.
- **Main credibility result.** The results survive exclusion of Illinois, remain visible within Illinois alone, and remain after controlling for local price variability, local housing growth, census-tract housing age, property type, and city status. Financing regressions also address land-trust reporting distortions.

- **Economic takeaway.** When information problems are severe, traders respond first by changing *who* trades with whom and *which* assets are brought to market. They respond much less through sophisticated financing design.

2 Data and measurement (key objects)

2.1 Why commercial real estate is a useful laboratory

- Commercial real estate is illiquid, heterogeneous, and highly local. These features make it hard for outsiders to value a property from publicly observed prices alone.
- Sellers are likely to know more than outsiders about neighborhood trends, local regulations, environmental risks, and the future income prospects of the property.
- The market is also useful for financing tests because loans are typically *nonrecourse*: lenders are secured by the property rather than by the owner's other assets.

Interpretation. The setting is attractive because information about the *property itself* and the *local market around it* matters a lot, and because financing contracts can be studied at the property level rather than at the firm level.

2.2 The COMPS database and sample construction

- The main data source is COMPS.com, a commercial real estate transaction database that collects information from buyers, sellers, and brokers and then cross-checks those reports.
- The raw sample contains 18,687 sales from January 1, 1996 to March 30, 1999.
- The main analysis keeps 10,351 transactions with recorded sale price, financing details, identities of principals, and property location.
- California transactions were available in an earlier version of the data, but the paper excludes California because the relevant assessment-ratio studies are not available there.

Interpretation. The paper prefers a smaller sample with the institutional variables needed to measure information quality cleanly.

2.3 Property, location, and participant variables

- COMPS records the location of the property and the city and state of the buyer and seller.
- Buyer and seller locations are matched to latitude and longitude coordinates, and distance from the property is computed in kilometers.
- Properties are grouped into apartments, vacant land, and commercial/industrial buildings. The paper also flags planned development deals.
- The data identify whether the buyer or seller is a professional broker trading on his own account.

Interpretation. Distance proxies for informational disadvantage, age proxies for the amount of income history available, and broker status proxies for being a known informed trader.

2.4 Financing variables

Each transaction can be financed with a combination of:

- cash,
- vendor-to-buyer financing (VTB),
- a new mortgage from a bank or other institution,
- and an assumed existing mortgage.

Key descriptive facts:

- VTB appears in 13.0% of transactions and covers 62.1% of price when used.
- New mortgages appear in 54.7% of transactions and cover 76.0% of price when used.
- Assumed mortgages appear in only 2.0% of transactions and cover 71.6% of price when used.
- Mean loan-to-value is 75.2%.

The paper also notes an institutional detail:

- VTB debt is usually junior to bank debt. Conditional on a second trust deed being present, VTB appears much more often than bank debt.

Interpretation. Seller financing exists in meaningful volume, but it is not obviously the clean signaling instrument that simple theory might suggest because the seller often retains a relatively safe debt claim rather than the riskiest equity claim.

2.5 Indirect and direct information measures

The paper uses two kinds of information variables.

Indirect information measures

- buyer distance from the property,
- property age,
- whether the buyer or seller is a broker.

Direct information measure

- the quality of public property-tax assessments in the relevant jurisdiction and property type.

Assessment quality is summarized by the coefficient of dispersion (COD):

$$\text{COD} = \frac{1}{N} \sum_{i=1}^N \frac{|R_i - R_{\text{med}}|}{R_{\text{med}}} \times 100, \quad (1)$$

where R_i is the assessment-to-market-value ratio for property i , and R_{med} is the median ratio in the jurisdiction-property-type cell.

Important implementation details:

- COD is taken from state assessment ratio studies, usually from 1998 or 1999, or the most recent available study.
- COD is matched at the level of assessing jurisdiction and broad property type.
- Higher COD means public assessments are noisier and therefore less informative.
- The paper also creates a **High COD** indicator for observations above the median COD of 13.31.

Interpretation. The paper is not using the assessment level itself. It uses the *dispersion* of assessment errors as a measure of how informative public assessments are for outsiders.

2.6 Dependent variables

The paper studies several dependent variables, depending on the prediction being tested:

- buyer distance from the property;
- property age;
- whether the buyer is a broker;
- whether a financing instrument is used;
- the size of financing, conditional on being used;
- and capitalization rate, defined as income divided by price.

Interpretation. The paper moves from trading behavior to asset selection to financing design. This is why the evidence can distinguish between direct no-trade style responses and financing-based responses.

2.7 Descriptive facts

- Buyers are on average 231.92 km from the property, and sellers are 264.38 km away.
- The mean COD is 29.36, with a large standard deviation of 28.31.
- Mean sale price is \$2.759 million, and median sale price is \$656,000.
- Apartments make up 23.3% of the sample, vacant land 26.1%, and commercial/industrial buildings 50.6%.
- High-COD areas differ sharply from low-COD areas in raw data, which is why the paper relies heavily on regression controls and subsample checks rather than raw comparisons.

Interpretation. The descriptive tables show that the information environment covaries with many other local features. The identification strategy therefore depends on the COD measure being used together with strong controls and robustness exercises.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and theoretical predictions

The paper distinguishes two information problems:

- sellers know more about local market conditions around the property;
- sellers may also know more about the physical state of the property.

The paper mainly focuses on the first problem: local-market information.

Predictions about limited participation, selective offering, and market segmentation

- **Prediction 1.** Buyers should be primarily local.
- **Prediction 2.** Buyer distance should be shorter in high-information-asymmetry environments.
- **Prediction 3.** The Prediction 2 effect should be strongest for properties with short income histories, proxied by young age.
- **Prediction 4.** Young properties should be less likely to be sold in high-information-asymmetry environments.
- **Prediction 5.** Brokers trading on their own account should be more likely to sell to other brokers, especially when information asymmetry is high.

Predictions about financial structure

Model A: VTB and bank-debt substitution

- **Prediction 6.** More VTB financing when information asymmetry is high.
- **Prediction 7.** Less new bank financing when information asymmetry is high.
- **Prediction 8.** More VTB when the existing assumed mortgage is large.

Model B: separating signaling

- **Prediction 9.** More VTB financing when information asymmetry is high.
- **Prediction 10.** Properties with large existing mortgages should be sold less often when information asymmetry is high.
- **Prediction 11.** More VTB when the existing mortgage is large.
- **Prediction 12.** Sale price should rise with the seller's retained stake, especially when information asymmetry is high.

Interpretation. The paper is really testing two different margins. The first margin is whether traders avoid information problems directly by staying local and segmenting the market. The second is whether they solve the problem indirectly with financing contracts.

3.2 Limited-participation regression

The baseline distance regression is

$$\text{BuyerDistance}_i = \alpha + \beta_1 \text{COD}_i + \beta_2 \text{HighCOD}_i + \beta_3 (\text{COD}_i \times \text{Age}_i) + X_i' \gamma + \varepsilon_i, \quad (2)$$

where not all three COD terms appear simultaneously in every specification.

The controls X_i include:

- city, land, apartment, and planned-development dummies;
- age of the property;
- $\log(\text{Price})$ and $\log(\text{Price})^2$;
- local price variability σ_{local} , defined as the standard deviation of capitalization rates within a 10-mile radius, excluding the property itself;
- MSA housing-price growth from 1996–1999;
- median age of primary housing stock in the census tract.

Interpretation. COD is supposed to shift information quality, while the control set absorbs property type, local heterogeneity, and recent local price dynamics.

3.3 Selective-offering regression

To test whether older properties are more likely to be sold in high-COD places, the paper estimates

$$\text{Age}_i = \alpha + \beta \text{COD}_i + X_i' \gamma + \varepsilon_i, \quad (3)$$

excluding properties with recorded age zero when needed.

Interpretation. A positive β means the market disproportionately brings *older* and hence easier-to-evaluate properties to market when the local information environment is poor.

3.4 Market-segmentation regression

To test broker-to-broker trading, the paper estimates

$$\Pr(\text{BuyerBroker}_i = 1) = \Lambda(\alpha + \beta_1 \text{SellerBroker}_i + \beta_2 \text{COD}_i + \beta_3 \text{SellerBroker}_i \times \text{COD}_i + X_i' \gamma), \quad (4)$$

using both logit and the semiparametric Klein-Spady binary-response estimator.

Interpretation. A positive coefficient on SellerBroker_i means informed agents trade with each other. A positive interaction with COD means this segmentation is strongest when information asymmetry is severe.

3.5 Financing regressions

For whether a financing instrument is used, the paper estimates binary-response models:

$$\Pr(F_{ik} > 0) = \Lambda(\alpha + \beta \text{COD}_i + \delta Z_i + X_i' \gamma), \quad (5)$$

where F_{ik} can be:

- VTB share,
- new-mortgage share,
- assumed-mortgage share,
- VTB^* , defined as seller financing in excess of the assumed-mortgage amount,
- VTB conditional on a second trust deed,
- vendor-only financing.

For the size of financing conditional on use, the paper estimates Powell's truncated regression:

$$F_{ik} \mid F_{ik} > 0 = \alpha + \beta \text{COD}_i + \delta Z_i + X_i' \gamma + u_i. \quad (6)$$

Interpretation. The logit regressions ask whether information affects *the extensive margin* of financing use. The truncated regressions ask whether it affects *the intensive margin* once that form of financing is chosen.

3.6 Price regression for the signaling test

To distinguish financing-substitution from signaling, the paper regresses capitalization rate on seller financing:

$$\text{CapRate}_i = \alpha + \beta \mathbf{1}\{\text{VTB}_i > 0\} + \theta \text{COD}_i + \lambda \mathbf{1}\{\text{VTB}_i > 0\} \times \text{COD}_i + X_i' \gamma + \varepsilon_i. \quad (7)$$

Because $\text{CapRate} = \text{Income}/\text{Price}$, a signaling model predicts a *negative* relationship between cap rate and seller retention when information asymmetry is high.

Interpretation. If VTB is a credible positive signal, buyers should pay more, which means lower cap rates for properties with seller retention.

3.7 Estimation choices and standard errors

- OLS regressions report White-corrected standard errors clustered at the level at which COD is measured: assessing jurisdiction by property type.
- Logit and Klein-Spady binary-response models use outer-product-based standard errors.
- Truncated regressions use bootstrap standard errors with 250 simulations.
- The paper cannot include state fixed effects in the main COD regressions because most COD variation is interstate. Instead, it uses exclusion and within-state checks.

Interpretation. The paper’s empirical strategy is more about measurement and robustness than about one single clever estimator.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main information effect

The paper does not use an instrument in the standard IV sense. Identification comes from exogenous variation in assessment quality across locations and property types.

The logic is:

- assessments are public;
- their dispersion is measured in official ratio studies;
- those studies are based on assessments made before the sale;
- the COD therefore shifts the information environment faced by outside buyers.

Interpretation. COD is treated as a direct measure of information quality, not as a reduced-form proxy like firm size or analyst coverage.

4.2 Why COD is a plausible signal of local information quality

The paper gives three reasons:

- Real estate practitioners actually pay attention to public assessments.
- Assessments need not be “better” than private appraisals; they only need to add information.
- What matters is the *dispersion* of assessment errors, not whether the median assessment level is high or low.

Interpretation. If all properties in a county were assessed at exactly 50% of value, outsiders could still invert the assessment perfectly. What hurts outsiders is noisy assessments, which is exactly what high COD captures.

4.3 Controls for local heterogeneity and price growth

The paper is careful that COD might proxy for other local conditions. The main distance and age regressions therefore control for:

- local cap-rate dispersion σ_{local} ;
- MSA-level housing-price growth;
- census-tract housing-age composition;
- property type;

- city-center status and price.

Interpretation. These controls try to net out local property heterogeneity and recent local booms, so that COD is not merely capturing hot markets or unusual building mixes.

4.4 Illinois and state-level concerns

Illinois matters because:

- it has very high COD values;
- it has low buyer distance;
- and it has a reputation for corruption, which could make outsiders worry about tax treatment.

The paper therefore shows the distance results:

- in the full sample,
- excluding Illinois,
- and within Illinois alone.

It also reports that a Hausman test does not reject the random-effects formulation against fixed effects, with $p = 0.43$.

Interpretation. The result is not just an Illinois story. It shows up without Illinois and even inside Illinois.

4.5 Property-condition versus local-market information

A potential objection is that COD may capture the condition of buildings rather than the local market. The paper argues against this by emphasizing:

- the focus is on local market conditions;
- appraisals and inspections are unlikely to reveal neighborhood trends cheaply;
- the strongest distance effect appears for young properties, exactly where public information about recent income and prices is most limited.

Interpretation. The $\text{COD} \times \text{Age}$ result is important because it fits a local-information story better than a generic hidden-defects story.

4.6 Land-trust reporting distortions

Illinois and Virginia recognize land trusts. In those states, seller financing can be recorded as bank financing because the bank technically holds title.

This matters for financing tests:

- some binary-response regressions for VTB and new mortgages exclude Illinois and Virginia;
- size regressions keep more states when the amount of financing is still measured correctly conditional on VTB being observed.

Interpretation. The paper is careful not to over-interpret financing patterns that could simply be reporting artifacts.

4.7 What the paper can and cannot distinguish

- The paper can strongly distinguish direct no-trade style responses from financing-based responses.
- It cannot fully determine whether seller finance should be interpreted as pecking-order substitution or as signaling, because the evidence on both is weak.
- The data are static, so the paper cannot study the timing of financing decisions or the dynamic evolution of capital structure.
- Equity stakes are not observed in the data, which limits the interpretation of seller retention.

Interpretation. The paper is strongest on showing that asymmetric information matters and on documenting *how people avoid trading*. It is weaker on showing that financing contracts solve the problem.

4.8 Relation to the literature

Relative to earlier work, the paper contributes by:

- providing a direct test of asymmetric information using exogenous differences in public assessment quality;
- testing predictions about limited participation, selective offering, and market segmentation in one setting;
- and showing that static financing theories receive much less support than the no-trade implications of adverse selection.

Interpretation. The paper fits a broader empirical pattern in corporate finance: pecking-order and signaling theories often look weaker in data than the underlying asymmetric-information problem itself.

5 Tables and figures

5.1 Table 1: descriptive statistics and the information environment

Read:

- Table 1 summarizes the 10,351-property estimation sample.
- Buyers are 231.92 km away on average, sellers 264.38 km away, and the mean COD is 29.36.
- Mean sale price is \$2.759 million, median sale price \$656,000, and mean loan-to-value 75.2%.
- VTB financing appears in 13.0% of deals, new mortgages in 54.7%, and assumed mortgages in 2.0%.

- The table also shows that high-COD places are systematically different from low-COD places in raw data.

Economic message. The table sets up the core empirical problem: information quality, buyer distance, and financing all differ a lot across places, so careful regression design is essential.

Table 1
Descriptive statistics on the COMPS (U.S.) database

Panel A: Summary information and coefficients of dispersion (COD)									
	No. of sales	Distance (km) from						COD	
		Buyer	Seller	Develop	Apt.	Land	Comm. & Ind.	Mean	σ
Overall	10,351	231.92	264.38	6.8%	23.3%	26.1%	50.6%	29.36	28.31
City-center	5,297	225.09	254.93	7.3%	27.7%	23.7%	48.7%	34.91	33.15
Non-city	5,054	239.08	274.28	6.3%	18.8%	28.6%	52.6%	23.54	20.59
Small deals	5,179	113.66	188.10	5.3%	25.6%	26.2%	48.2%	33.41	29.07
Large deals	5,172	350.35	340.76	8.3%	21.0%	26.0%	53.0%	25.30	26.93
NV	1,603	360.06	312.74	7.5%	12.7%	54.0%	33.4%	5.82	1.11
MA	348	213.89	149.64	4.6%	72.1%	27.9%	0.0%	7.69	3.55
MD	854	267.50	304.50	3.0%	20.0%	3.3%	76.7%	9.62	5.17
VA	1,372	229.92	273.09	4.1%	6.5%	52.6%	41.0%	13.79	4.70
TX	1,311	370.90	463.99	16.5%	25.6%	21.7%	52.6%	7.43	1.24
IL	4,219	128.62	174.64	6.0%	32.3%	13.4%	54.3%	58.32	22.81
CO	644	273.58	315.82	2.3%	0.0%	21.6%	78.4%	13.92	0.90
High COD	5,155	149.88	200.06	5.3%	27.6%	13.5%	58.9%	50.84	26.10
Low COD	5,196	313.31	328.19	8.3%	19.0%	38.6%	42.3%	8.04	2.90

Panel B: Pricing and financing information									
	Sale price (\$,000)		VTB		New mortgage		Assumed mortgage		Loan/value
	Mean	Median	Freq(%)	%Price	Freq(%)	%Price	Freq(%)	%Price	
Overall	\$2,759	\$656	13.0%	62.1%	54.7%	76.0%	2.0%	71.6%	75.2%
City	\$2,831	\$613	14.8%	62.9%	58.5%	75.0%	2.4%	77.6%	75.1%
Non-city	\$2,683	\$713	11.1%	60.6%	50.8%	77.3%	1.7%	57.4%	75.3%
Small deals	\$407	\$390	14.9%	67.6%	56.9%	77.6%	0.6%	59.0%	77.4%
Large deals	\$5,114	\$1,520	11.1%	60.9%	52.5%	75.8%	3.5%	71.7%	75.0%
NV	\$2,224	\$790	32.4%	65.6%	43.4%	71.5%	4.2%	64.4%	71.2%
MA	\$2,687	\$658	18.4%	52.4%	58.0%	74.4%	4.3%	67.3%	73.5%
MD	\$2,735	\$700	20.1%	68.1%	48.7%	75.3%	1.1%	64.2%	74.8%
VA	\$4,140	\$894	8.7%	59.2%	36.3%	81.9%	0.7%	30.0%	77.6%
TX	\$3,512	\$970	13.9%	53.2%	60.4%	76.9%	4.6%	67.7%	75.2%
IL	\$2,382	\$530	4.1%	58.7%	66.5%	75.6%	0.8%	80.8%	76.7%
CO	\$2,150	\$625	18.2%	67.6%	39.6%	69.6%	2.5%	59.7%	71.2%
High COD	\$2,436	\$550	6.7%	60.8%	62.3%	75.7%	1.0%	78.9%	76.6%
Low COD	\$3,078	\$820	19.2%	62.4%	47.2%	76.2%	3.0%	62.0%	74.2%

Descriptive statistics on the COMPS commercial real estate transactions from the United States over the period January 1, 1996, to March 30, 1999, are reported above. Panel A reports general statistics on the number of sales, average distance buyers and sellers are from the property, and the percentage of property sales that are planned development (Develop), apartments (Apt.), vacant land (Land), and commercial and industrial buildings (Comm. & Ind.). The last two columns report the mean and standard deviation (σ) of the coefficient of dispersion (COD), which is the dispersion around the median sale price-to-assessment value ratio within a region and for a particular property type. These figures are obtained from state property assessment ratio studies. COD measures are provided for each assessing jurisdiction

Table 2
Limited participation and selective offering

	Dependent variable = buyer distance from property (km)								
	All	Exclude IL	IL only	All	Exclude IL	IL only	All	Exclude IL	IL only
No. of observations	9805	5776	4029	9805	5776	4029	6647	3304	3343
City	47.60* (1.89)	70.34** (2.26)	-38.75** (-2.42)	30.93 (1.40)	71.52** (2.88)	-72.77** (-3.50)	31.93 (1.26)	58.70** (2.06)	-26.80** (-2.67)
Age	-2.00** (-4.30)	-2.26** (-4.12)	-0.49** (-2.62)	-1.75** (-4.26)	-2.25** (-4.10)	-0.37** (-2.27)	-3.51** (-5.46)	-3.46** (-2.56)	-5.45** (-3.29)
Land	-62.99** (-2.61)	-101.89** (-5.93)	108.48** (3.00)	-80.72** (-3.74)	-99.66** (-6.53)	33.12 (1.08)			
Apartment	40.62 (0.88)	95.87* (1.72)	-28.80** (-2.98)	34.56 (0.85)	95.12* (1.73)	-21.80 (-1.38)	55.49 (1.46)	122.91** (2.47)	-17.54** (-2.20)
Develop	-42.97 (-1.34)	-36.03 (-0.77)	-52.38* (-1.75)	-52.62* (-1.70)	-35.52 (-0.75)	-53.07* (-1.79)	-26.29 (-0.70)	-18.54 (-0.33)	-11.35 (-0.35)
σ_{local}	-6.68 (-0.20)	29.16 (0.67)	23.75 (0.57)	8.74 (-0.84)	20.61 (0.26)	-20.65 (-0.78)	-26.69 (-0.60)	9.07 (-0.74)	0.40 (0.40)
log(Price)	-484.6** (-3.44)	-547.4** (-3.04)	-404.7** (-3.01)	-489.6** (-3.47)	-548.4** (-3.05)	-407.0** (-3.02)	-420.9** (-3.12)	-398.7** (-2.07)	-401.8** (-2.61)
(log(Price)) ²	21.51** (4.20)	24.43** (3.82)	16.97** (3.83)	21.65** (4.22)	24.46** (3.83)	17.04** (3.84)	19.51** (4.10)	19.77** (2.97)	17.06** (3.37)
CTA age	-0.40 (-0.67)	-1.60* (-1.92)	-0.76 (-1.31)	-1.81** (-2.13)	0.06 (0.50)	-1.81** (-2.13)	0.09 (1.09)	-1.46 (-0.65)	0.85** (-0.71)
MSA growth	194.49 (0.50)	34.14 (0.09)	0.52 (1.42)	264.89 (0.78)	60.48 (0.19)	0.41 (1.08)	-276.56 (-0.65)	-316.53 (-0.71)	
COD	-1.48** (-3.08)	-6.13** (-2.83)	-1.67** (-2.96)				-5.13** (-5.16)	-11.48** (-2.10)	-4.67** (-4.64)
COD $\times$ age							0.05** (4.07)	0.13 (1.17)	0.09** (2.90)
High COD				-92.96** (-4.29)	-63.01** (-4.02)	-118.00** (-8.50)			

Results from the regression of buyer distance from the property (in km) on a set of control variables and two direct measures of exogenous information, COD and HighCOD, are reported over the period January 1, 1996, to March 30, 1999. COD is the coefficient of dispersion measure and HighCOD is a dummy variable equal to one for all properties with associated COD measures above the median measure (13.31). The control variables include city, land, apartment, and planned development dummies, the age of the property, the log and squared log of the sale price, a measure of local price variability, which is the standard deviation of commercial property capitalization rates within a 10 mile radius, excluding the property itself, σ_{local} , the average growth rate on the index of all housing sales within each municipal statistical area (MSA) over the 1996-1999 period, provided by Fannie Mae and Freddie Mac, and the median age of primary housing stock in each census tract, provided by the U.S. Census Bureau. Each property is identified with its MSA and census tract via its latitude and longitude coordinates. An interaction term between COD and age is also employed. Regressions are run for the whole sample, excluding properties from the state of Illinois, and for all properties within the state of Illinois only. The regressions are estimated via ordinary least squares (OLS), with *t*-statistics reported in parentheses using White (1980)-corrected standard errors that account for group-wise clustering at the

5.2 Table 2: limited participation and local-buyer effects

Read:

- Table 2 regresses buyer distance on COD, High COD, and COD interacted with age.
- The COD coefficient is negative in all samples: full sample, excluding Illinois, and Illinois only.
- The High COD dummy implies buyers in high-COD locations are markedly closer to the property.
- The interaction between COD and age is positive, meaning the local-buyer effect is strongest for younger properties.

Economic message. The core no-trade prediction works very well. When outsiders are at a bigger informational disadvantage, they pull back and the buyer pool becomes more local.

5.3 Table 3: selective offering and market segmentation

Read:

- Panel A of Table 3 shows a positive relationship between property age and COD.

- That means older properties, with longer histories, are more likely to be sold where information asymmetry is worse.
- Panel B shows that sellers who are brokers are much more likely to sell to buyers who are brokers.
- The seller-broker effect becomes stronger when interacted with COD in the full sample and in the sample excluding Illinois.

Economic message. High-information-asymmetry environments reshape *what* gets sold and *who* trades with whom. The market becomes more segmented and easier-to-evaluate assets dominate transactions.

Table 3
Selective offering and market segmentation

Dependent variable =	Panel A: Property age			Panel B: Buyer is a broker			
	All	Exclude IL	IL only	All	All	Exclude IL	IL only
No. of observations	6557	3225	3332	7468	7468	4459	4013
Regression model	OLS	OLS	OLS	Logit	Klein-Spady	Logit	Logit
City	10.061** (2.63)	-1.828 (-0.72)	26.947** (12.81)	0.829** (6.34)	0.128 (0.20)	0.873** (5.31)	0.322 (1.07)
Age				0.002 (0.75)	-0.003 (-0.26)	0.002 (0.42)	0.010** (2.55)
Land				0.389** (2.40)	-0.232 (-0.29)	0.458** (2.24)	0.610 (1.24)
Apartment	7.671** (2.22)	8.551** (2.19)	6.398** (9.92)	-0.329** (-2.11)	-0.141 (-0.18)	0.343 (1.61)	0.555** (2.76)
Develop	-1.442 (-1.19)	-1.763 (-0.96)	-1.688 (-1.08)	-0.130 (-0.57)	-0.015 (-0.03)	-0.014 (-0.05)	-0.136 (-0.35)
σ_{local}	1.857 (0.61)	1.205 (0.47)	0.546 (0.74)	0.073 (0.38)	-0.018 (-0.02)	0.003 (0.01)	-0.469 (-1.18)
Seller is a broker				1.324** (6.57)	2.416** (3.52)	-0.060 (-0.69)	4.067** (6.24)
(Seller is a broker) $\times$ COD				0.022** (4.20)	0.147** (2.24)	1.424** (2.42)	-0.012 (-1.16)
log(Price)	-11.464 (-1.18)	-23.255** (-3.83)	9.116 (0.81)	-0.051 (-0.06)	-1.000 (-0.74)	-0.846 (-0.74)	-0.342 (-0.27)
(log(Price)) ²	0.254 (0.77)	0.632** (3.27)	-0.449 (-1.15)	0.001 (0.01)	-0.023** (-2.48)	0.026 (0.67)	0.013 (0.30)
Buyer distance				-0.001 (-0.76)	0.002 (0.39)	-0.004** (-2.81)	0.001 (0.05)
CT age	0.466** (3.15)	0.484** (3.80)	0.272 (1.33)	0.007 (1.41)	0.001 (0.03)	0.003 (0.45)	-0.013* (-1.82)
MSA growth	103.182** (1.98)	132.658** (3.71)		-6.776** (-2.78)	-0.353 (-0.03)	-1.876 (-0.63)	
COD	0.335** (2.42)	0.758** (2.32)	0.171** (2.56)	-0.007** (-2.14)	0.003 (0.13)	-0.103** (-3.68)	0.004 (0.44)

Panel A reports results from the regression of property age (excluding properties with recorded ages of zero, which include all vacant land sales) on the direct information measure COD over the period January 1, 1996, to March 30, 1999. The control variables include city, apartment, and planned development dummies, the log and squared log of the sale price, a measure of local price variability, which is the standard deviation of commercial property capitalization rates within a 10 mile radius, excluding the property itself, σ_{local} , the average growth rate on the index of all housing sales within each municipal statistical area (MSA) over the 1996 to 1999 period, provided by Fannie Mae and Freddie Mac, and the median age of primary housing stock in each census tract, provided by the U.S. Census Bureau. The regressions are estimated via ordinary least squares (OLS), with *t*-statistics reported in parentheses using White (1980)-corrected standard errors that account for group-wise clustering at the level in which COD is measured (assessing jurisdiction and property type). Panel B reports binary response regression results where the dependent variable is whether the buyer is a broker trading on his own account, estimated via logit and the Klein and Spady (1993) semiparametric model with *t*-statistics reported in parentheses computed from the outer-product matrix

5.4 Table 4: financing tests

Read:

- Table 4 is the main financing table and is split into a binary-response panel and a truncated-regression panel.
- The COD coefficient is weakly positive for the frequency of VTB financing, but there is essentially no support for the size of VTB financing increasing with COD.
- New bank financing does not fall in high-COD places; in the truncated regression its size even rises with COD, opposite to theory.
- Vendor-only financing shows some positive frequency response to COD, but most other financing predictions are rejected.

Economic message. The financing evidence is the weak part of the paper. Adverse selection clearly affects trading patterns, but it does not translate into consistent support for static capital-structure theories.

Table 4
Is financial structure affected by information asymmetry?

Dependent variable =	Panel A: binary response (Logit)			Panel B: truncated regression (Powell (1986))						
	VTB	New mortgage	VTB*	VTB 2nd trust	Vendor only	VTB	New mortgage	Assumed mortgage	VTB 2nd trust	Vendor only
No. of observations	2260	2260	121	755	2260	898	1523	156	297	615
City	0.1542 (1.45)	-0.3355** (-3.24)	-0.3016 (-0.36)	-0.1483 (-0.33)	0.2511** (2.24)	-0.0131 (-0.44)	-0.0381** (-3.32)	-0.0533** (-1.73)	0.0790 (0.03)	-0.0460** (-2.47)
Age	0.0159** (6.00)	-0.0065** (-2.44)	0.0179 (0.73)	0.0261** (2.44)	0.0137** (4.86)	-0.0015 (-1.49)	-0.0001 (-0.01)	-0.0002 (-0.19)	-0.0208 (-0.55)	0.0006 (0.98)
Land	1.5353** (10.19)	-1.1399** (-7.77)	1.9061* (1.68)	0.0108 (0.02)	1.5572** (10.02)	-0.1571 (-1.46)	-0.0985** (-6.64)	-0.1324** (-2.35)	-6.3572 (-1.62)	-0.0756** (-2.65)
Apartment	-0.0231 (-0.17)	-0.0253 (-0.19)	-0.2174 (-0.24)	-0.1257 (0.22)	-0.1592 (-1.07)	-0.0894 (-1.11)	0.0267** (2.45)	0.0696** (2.71)	-0.3297 (-0.22)	0.0130 (0.54)
Develop	-0.2123 (-1.20)	-0.0169 (-0.10)	-1.4493 (-0.47)	-0.4456 (-0.49)	-0.0366 (-0.20)	0.0140 (0.22)	-0.0048 (-0.30)	-0.0119 (-0.18)	1.0848 (0.65)	0.0413 (1.12)
σ_{local}	0.0199 (0.13)	0.1086 (0.75)	-0.1140 (-0.11)	-0.2150 (-0.39)	0.0871 (0.55)	-0.0207 (-0.45)	0.0394** (3.21)	0.0161 (0.33)	0.1351 (0.09)	0.0124 (0.41)
Broker	-0.5709** (-5.56)	0.5343** (5.29)	-1.0206 (-1.28)	-0.7104* (-1.69)	-0.5779** (-5.42)	-0.0617** (-2.11)	-0.0191 (-1.45)	0.0426 (1.13)	0.9847 (0.37)	-0.0524** (-3.16)
Seller is a broker	0.0202 (0.11)	0.1350 (0.72)	-4.4502 (-0.75)	-1.3361 (-0.90)	-0.1330 (-0.67)	-0.1120** (-2.47)	-0.0176 (-0.82)	-0.0225 (-0.39)	-6.5476* (-1.84)	-0.1002** (-2.60)
Buyer distance	-0.0003** (-3.63)	0.0001 (0.42)	-0.0002 (-0.44)	0.0004* (1.87)	-0.0004** (-3.70)	-0.0001 (-0.59)	-0.0001 (-0.91)	-0.0001 (-1.29)	0.0022 (0.95)	-0.0001 (-0.64)
CT age	0.0003 (0.07)	0.0001 (0.02)	0.0267 (0.71)	-0.0006 (-0.03)	-0.0034 (-0.70)	-0.0014 (-0.72)	-0.0002 (-0.41)	0.0010 (0.60)	0.0168 (0.35)	-0.0002 (-0.23)

(a) Table 4, first page

Table 4
Continued

Dependent variable =	Panel A: binary response (Logit)			Panel B: truncated regression (Powell (1986))						
	VTB	New mortgage	VTB*	VTB 2nd trust	Vendor only	VTB	New mortgage	Assumed mortgage	VTB 2nd trust	Vendor only
MSA growth	-10.7492** (-5.47)	7.5166** (3.90)	1.8195 (0.11)(0.14)	1.1812 (-5.46)	-11.4205** (-5.46)	0.6475 (0.65)	0.0594 (0.33)	-0.2425 (-0.54)	8.6431** (2.24)	0.0466 (0.10)
COD	0.0355* (1.69)	-0.0067 (-0.32)	-0.0438 (-0.20)	-0.0958 (-0.84)	0.0440** (2.01)	-0.0008 (-0.63)	0.0042** (2.05)	0.0005 (0.30)	0.0125 (0.10)	0.0033 (0.65)
Assumed			-3.0678 (-0.85)							

Results from the regression of various financing types on direct and indirect information variables plus property, buyer, seller, and location characteristics are reported above over the period January 1, 1996, to March 30, 1999. Six sets of dependent variables are used: vendor-to-buyer financing (VTB) scaled by sale price, new mortgage scaled by sale price, assumed mortgage scaled by sale price, VTB scaled by sale price in excess of the amount of assumed mortgage (VTB*), VTB financing conditional on a second trust deed being used, and vendor only financing (e.g., financed solely by the seller). Panel A reports coefficient estimates under logit, where the dependent variable is one if the financing type is used, and zero otherwise. Panel B reports coefficient estimates under the truncated regression model of Powell (1986), where the data is truncated to only those observations where the dependent variable (financing type) is positive. *T*-statistics are reported in parentheses, with standard errors calculated via the outer-product matrix for the binary response model, and via bootstrapping (250 simulations) for the truncated regression model. Binary response regressions of VTB and new mortgages as well as the truncated regression of new mortgages, exclude states that recognize land trusts to control for distorted reporting effects arising from land trust deals. The regressors include COD, city, land, apartment, and planned development dummies, the age of the property, a measure of local price variability, which is the standard deviation of commercial property capitalization rates within a 10 mile radius, excluding the property itself, σ_{local} , dummy variables for the presence of a broker and whether the seller is a broker, buyer distance from the property, the average growth rate on the index of all housing sales within each municipal statistical area (MSA) over the 1996–1999 period, provided by Fannie Mae and Freddie Mac, and the median age of primary housing stock in each census tract, provided by the U.S. Census Bureau. The constant term is not reported for brevity.

**Indicates significance at the 10% and 5% levels, respectively.

(b) Table 4, continuation

5.5 Table 5: does seller retention affect the price?

Read:

- Table 5 regresses capitalization rate on the presence of seller financing, COD, and their interaction.
- The coefficient on seller financing is not statistically significant.

- The interaction between seller financing and COD is also not significant.
- The result remains weak even when the sample is restricted to the most distant buyers or the youngest properties.

Economic message. There is no meaningful support for the signaling prediction that seller retention raises price when information asymmetry is high.

Table 5
Do financial decisions influence the price?

	Dependent Variable: Capitalization rate = income/price			
	All properties	Most distant	Youngest	
No. of observations	744	744	407	163
City	-0.4230** (-2.02)	-0.4210** (-2.02)	-0.5776** (-1.96)	-0.1759 (-0.89)
Age	0.0171** (2.29)	0.0169** (2.34)	0.0194** (2.16)	0.0808** (2.80)
Land	0.5190 (1.19)	0.4942 (1.16)	0.5310 (1.10)	0.8448** (1.91)
Apartment	0.1094 (0.43)	0.0965 (0.39)	0.2472 (0.87)	-0.6683** (-3.08)
Develop	-0.0416 (-0.23)	-0.0430 (-0.24)	-0.3681** (-4.77)	-0.1644 (-1.24)
σ_{local}	0.9315** (2.63)	0.9378** (2.70)	1.3739** (3.07)	0.1034 (0.36)
Broker	0.3252 (1.02)	0.3227 (1.00)	0.1142 (0.40)	0.5013 (1.20)
Seller is a broker	-0.4071 (-0.78)	-0.4401 (-0.84)	-0.1195 (-0.45)	-0.0425 (-0.17)
Buyer distance	-0.0001 (-1.50)	-0.0001* (-1.80)	0.0001 (1.51)	-0.0001** (-2.37)
CT age	0.0172** (1.99)	0.0170** (1.98)	0.0233* (1.81)	0.0158 (1.27)
MSA growth	-7.0595 (-1.61)	-7.1565* (-1.73)	-7.5520 (-1.28)	-2.1928 (-0.37)
COD	0.0060 (0.11)	-0.0249 (-0.47)	-0.0074 (-0.11)	0.0659 (1.65)
VTB > 0	-0.0826 (-0.22)	-0.9597 (-1.07)	0.5332 (1.29)	-0.1194 (-0.23)
(VTB > 0) × COD		0.1067 (1.20)		

Results from the regression of property capitalization rates (net income on the property divided by the sale price) on the presence of the seller's retained stake in the property (a dummy variable equal to one if seller financing is present) plus a set of controls for broker usage, buyer, seller, and property characteristics, including city, land, apartment, and planned development dummies, the age of the property, a measure of local price variability, which is the standard deviation of commercial property capitalization rates within a 10 mile radius, excluding the property itself, σ_{local} , dummy variables for the presence of a broker and whether the seller is a broker, buyer distance from the property, the average growth rate on the index of all housing sales within each municipal statistical area (MSA) over the 1996–1999 period, provided by Fannie Mae and Freddie Mac, and the median age of primary housing stock in each census tract, provided by the U.S. Census Bureau are reported above over the period January 1, 1996, to March 30, 1999. Also included as regressors are COD and an interaction between the seller's retained stake and COD. Regressions are run over all properties for which cap rates are available, for properties with the most distant half of buyers, and for the youngest properties (less than five years of income history). Coefficient estimates are calculated via ordinary least squares (OLS), with *t*-statistics reported in parentheses, where standard errors are White (1980)-corrected and account for groupwise clustering at the level in which COD is

5.6 Table 6: summary of predictions and their verification

Read:

- Table 6 summarizes the paper's full scorecard.
- Predictions 1–5, covering local buyers, selective offering, and broker segmentation, are all verified.
- Predictions 6–12, covering VTB, bank debt, signaling, and seller retention, are mostly rejected or only weakly supported.
- The only financing result with some support is the higher probability of vendor-only financing in high-information-asymmetry environments.

Economic message. The summary table is the paper in one page: asymmetric information matters a lot, but it is mostly handled through direct market-structure responses rather than through financial engineering.

Table 6
Summary of theoretical predictions and their verification in the data

Prediction	Table, Column	Verified?
Limited participation, selective offering, and market segmentation		
1. <i>Buyers are local.</i>	1	Yes
2. <i>Buyer distance is smaller when information asymmetry is high (e.g., when COD is high).</i>	2, 1-6	Yes
3. <i>Buyer distance is smallest when information asymmetry is high and the property is young.</i>	2, 7-9	Yes
4. <i>Young properties are less likely to be sold when information asymmetry is high.</i>	3, 1-3	Yes
5. <i>Brokers are more likely to sell to other brokers, particularly when information asymmetry is high.</i>	3, 4-7	Yes
Financial structure		
Adverse selection model A: seller and bank debt substitution		
6. <i>More seller financing (VTB) when information asymmetry is high.</i>		
Frequency (%)	4, 1	Weak
Magnitude (\$)	4, 6	No
2nd Trust only (%)	4, 4	No
2nd Trust only (\$)	4, 9	No
Vendor only (%)	4, 5	Yes
Vendor only (\$)	4, 10	No
7. <i>Less bank financing when information asymmetry is high.</i>		
Frequency (%)	4, 2	No
Magnitude (\$)	4, 7	No
8. <i>More VTB when existing (assumed) mortgage is large.</i>	4, 3	No
Adverse selection model B: separating signaling		
9. <i>More VTB when information asymmetry is high. (same as 6)</i>	See above	See above
10. <i>Less assumed mortgage when information asymmetry is high.</i>	4, 8	No
11. <i>More VTB when existing mortgage is large. (same as 8)</i>	See above	See above
12. <i>Sale price positively related to seller's retained stake (VTB), particularly when information asymmetry is high.</i>	5, 1-2	No
Most distant buyers only	5, 3	No
Youngest properties only	5, 4	No

with our results. The timing of financing decisions and the evolution of capital structure, however, may be more affected by information consid-

6 Short summary

- The paper studies asymmetric information in U.S. commercial real estate using a direct measure of local information quality based on public-assessment accuracy.
- High COD means poor assessments and therefore a worse information environment for outsiders.
- Buyers become much more local in high-COD areas, especially for younger properties.
- Older properties are more likely to be sold in those environments, consistent with selective offering.
- Brokers are more likely to trade with brokers, especially where COD is high, which indicates market segmentation between informed and uninformed traders.
- The financing evidence is weak: seller financing rises only weakly in frequency, bank financing does not fall as predicted, and seller retention does not raise prices.
- The main lesson is that asymmetric information changes *participation*, *asset selection*, and *counterparty matching* more than it changes static financing design.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that asymmetric information is economically important in commercial real estate markets. It does so with an unusually direct measure of the information environment: the quality of local property-tax assessments. Where public assessments are noisy, buyers stay local, younger properties are less likely to come to market, and informed brokers trade more with other informed brokers.

7.2 Economic intuition

If outsiders cannot reliably infer local market value, they do not simply accept the information disadvantage and compensate with price. Instead, they either stay away, buy only easier-to-evaluate properties, or leave trading to people who are already informed. These are exactly the no-trade and selective-offering implications of adverse-selection models.

7.3 Takeaway

The paper's most important message is that the market solves information problems first through *who trades* and *what gets traded*, and only weakly through *how the trade is financed*. That makes the no-trade logic of asymmetric information look more empirically powerful than the static capital-structure logic in this setting.